Module 1: Foundations, Models, Physical and Data Link Layers

Data communication, OSI and TCP/IP, transmission media, switching, multiplexing, line coding and PCM, error detection and correction, and the MAC sublayer

Exam: Wednesday 18 November 2026

Exam Focus

Scope. Syllabus 1.1 to 1.4: overview of data communication and networking; network models (OSI and TCP/IP); the physical layer (transmission media, switching, multiplexing); and the data link layer (error detection and correction, data link control).

Module 1 carries CO1 and dominates the CAE-1 paper. In the six papers we have, Module 1 accounts for **more marks than any other module**: every CAE-1 is almost entirely Module 1. Expect the layered-approach question, the OSI versus TCP/IP comparison, transmission media, multiplexing, switching, and a CRC calculation.

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Data communication

Definition and components

Data communication is the exchange of data between two devices through some transmission medium, in which the data is represented as agreed patterns of electromagnetic signals.

Past-paper question

Sept 2026 CAE-1 Q1B(a). Define Data Communication and Networking. Explain the five basic components of a data communication system.

Solution

A data communication system has **five** components. Learn these five and the order, because the question is worth 10 marks and is pure recall.

Component	Also called	Role
Message	Data	The information to be communicated: text, numbers, images, audio, video.
Sender	Source	The device that sends the message: computer, phone, camera.
Receiver	Destination	The device that receives the message.
Transmission medium	Channel	The physical path the message travels: twisted pair, coaxial cable, fibre, radio waves.
Protocol	Rules	The set of rules governing communication: what is sent, how, and when. Without a protocol two devices may be connected but cannot communicate.

Networking is the interconnection of two or more devices so that they can share resources and exchange data. A **protocol** is the agreement that makes the exchange meaningful: syntax (format), semantics (meaning), and timing (when to send and how fast).

Performance characteristics

Past-paper question

Sept 2026 CAE-1 Q1B(b). Explain the important network performance characteristics: Bandwidth, Data Rate, Throughput, Latency, and Jitter.

Solution

Term	Meaning	Units and note
Bandwidth	The *capacity* of a medium to carry data: the range of frequencies it can pass. In digital terms, the maximum bit rate a link can support.	Hertz (analogue); bits per second (digital).
Data rate	The actual number of bits transmitted per second by a device.	bps, kbps, Mbps, Gbps.
Throughput	The *actual* rate of successful data delivery, after protocol overhead, retransmissions and congestion.	Always $\leq$ bandwidth.
Latency	Total time for a message to travel from sender to receiver: propagation + transmission + queueing + processing.	Milliseconds.
Jitter	Variation in latency between successive packets.	Milliseconds. Fatal for real-time voice and video.

Bandwidth versus throughput is the distinction most often asked. Bandwidth is the pipe's size; throughput is what actually flows through it. A 100 Mbps link delivering 60 Mbps of useful data has a throughput of 60 Mbps.

Latency versus jitter. Latency is how late a packet is. Jitter is how *unevenly late* successive packets are. A voice call tolerates a steady 150 ms delay; it does not tolerate 20 ms, 300 ms, 30 ms, 400 ms.

Do Not Confuse

Bandwidth and data rate are sometimes used interchangeably, and sometimes not. In strict usage, bandwidth is the frequency range of the medium, and data rate is the bit rate achieved. In the Shannon and Nyquist formulas the two are linked but distinct. If a question uses them loosely, define both terms in your answer and state the relation, and you will not lose marks either way.

Transmission modes

Mode	Description
Simplex	One direction only, always. Keyboard to computer; radio broadcast.
Half duplex	Both directions, but only one at a time. Walkie-talkie.
Full duplex	Both directions simultaneously. Telephone; a normal switched Ethernet link.

Network types and topologies

By geographical scope

Type	Scope	Characteristics
PAN	A few metres	Personal area network: Bluetooth, USB, a phone tethered to a laptop.
LAN	A building or campus	High data rate, low cost, privately owned. Ethernet and Wi-Fi.
MAN	A city	Connects several LANs across a city; usually owned by a provider or a municipal body.
WAN	Country or global	Connects LANs and MANs over long distance; uses leased lines, MPLS, satellite. The Internet is the largest WAN.

By topology

Topology	How it works	Strength / weakness
Bus	All nodes tap one shared backbone cable.	Cheapest cabling; a break in the backbone kills the network; collisions.
Star	All nodes connect to a central hub or switch.	Easy to diagnose, one cable per node; the centre is a single point of failure.
Ring	Each node connects to two neighbours, forming a loop.	Predictable access (token ring); a break can isolate nodes.
Mesh	Every node connects to every other (full) or to several (partial).	Fault tolerant, many paths; expensive cabling, $n(n-1)/2$ links for full mesh.
Tree	A hierarchy of star-connected segments.	Scales to large sites; the root is critical.
Hybrid	A combination, as in most real networks.	Flexible; complex to manage.

Remember

For a **full mesh** of n nodes the number of links is $\dfrac{n(n-1)}{2}$ and each node needs $n-1$ interfaces. For 10 nodes that is 45 links. This arithmetic is a common 3-mark question.

Network models

Why layer the network at all

Past-paper question

Sept 2026 CAE-1 Q1A. Explain the importance of the layered approach in computer networking. Discuss how layering simplifies network design, implementation, troubleshooting, and interoperability, with suitable examples.

Solution

Layering divides the complex job of moving data between applications on different machines into a stack of self-contained functions, where each layer uses the services of the layer below and provides services to the layer above.

1. **Divide and conquer.** Each layer solves one well-defined problem. The designer of the transport layer does not need to know whether the link is fibre or radio.
2. **Abstraction and independent implementation.** A layer's internals can be replaced without touching its neighbours. Ethernet can be swapped for Wi-Fi underneath IP and IP does not change.
3. **Interoperability through standard interfaces.** Because the boundary between layer n and layer $n+1$ is standardised, a vendor can build a TCP stack and another vendor an Ethernet card, and they work together. This is what makes multi-vendor networking possible.
4. **Modular troubleshooting.** A failure can be localised to a layer. "The cable is fine (layer 1), the switch sees the MAC (layer 2), but no ping reply (layer 3)" is a systematic diagnosis. That is exactly what `ping`, `traceroute` and packet captures exploit.
5. **Reuse and specialisation.** The same transport layer serves web, mail and file transfer. Each new application does not re-implement reliability.
6. **Easier evolution.** IPv6 could be introduced at layer 3 while layers 1, 2 and 4 stayed as they were.

The cost: layering adds header overhead and some duplication of function (error checking appears at several layers), and a strictly layered design can be slower than a monolithic one.

The OSI reference model

Seven layers, bottom to top. Learn the function *and* the PDU *and* the addressing used at each layer.

No.	Layer	PDU	Main function
7	Application	Data	Provides network services to the user: HTTP, FTP, SMTP, DNS.
6	Presentation	Data	Translation, compression, encryption: character encoding, TLS.
5	Session	Data	Establishes, manages and terminates sessions; synchronisation and checkpointing.
4	Transport	Segment / Datagram	End-to-end delivery, segmentation and reassembly, flow and error control, port addressing. TCP and UDP.
3	Network	Packet	Logical addressing and routing between networks. IP, ICMP, routing protocols.
2	Data link	Frame	Framing, physical (MAC) addressing, error detection, flow control, media access. Ethernet, PPP.
1	Physical	Bit	Transmission of raw bits over a medium: signalling, voltages, connectors, bit rate.

Remember

Mnemonic, bottom to top: *Please Do Not Throw Sausage Pizza Away* = Physical, Data link, Network, Transport, Session, Presentation, Application.

Which layer, quickly:

- "Which route?" → network (3)
- "Which port?" → transport (4)
- "Which MAC address?" → data link (2)

- "Which voltage, which pin?" → physical (1)
- "Encrypt, compress, convert character set" → presentation (6)

The TCP/IP model

Past-paper question

Sept 2026 CAE-1 Q3B. Explain the TCP/IP protocol suite. Discuss its layered architecture and compare it with the OSI reference model.

Solution

The TCP/IP suite has **four** layers in the classic form, or five if the physical layer is counted separately.

TCP/IP layer	Function	Protocols
Application	Everything the user sees; combines OSI 5, 6 and 7	HTTP, FTP, SMTP, DNS, DHCP, SNMP, TLS
Transport	End-to-end delivery, reliability, flow and congestion control	TCP, UDP, SCTP
Internet	Logical addressing and routing	IP, ICMP, IGMP, ARP (at the boundary), OSPF, BGP
Network access (link)	Framing, MAC addressing, media access, bit transmission	Ethernet, Wi-Fi, PPP

Comparison with OSI:

Basis	OSI	TCP/IP
Layers	7	4
Nature	A theoretical reference model, model first then protocols	A practical model, protocols first then the description
Session and presentation	Separate layers	Folded into the application layer
Guaranteed delivery	Defined at the transport and network layers	Only the transport layer (TCP) guarantees it
Use	Teaching and diagnosis	Actual implementation on every host
Design origin	ISO, top-down	ARPANET/DoD, bottom-up

Why TCP/IP won: it was implemented and shipped before OSI was finished, it was free and open, and it was already running the Internet by the time OSI was standardised.

Encapsulation and decapsulation

Past-paper question

Sept 2024 CAE-1 Q2. Write notes on: 1. Multiplexing 2. Hierarchical Layering 3. Encapsulation/Decapsulation 4. PDU.

Solution

Encapsulation is the process of adding a header (and sometimes a trailer) at each layer as data moves *down* the stack.

Layer	What is added
Application	The user's message (e.g. an HTTP request)
Transport	TCP header with source and destination port, sequence number, checksum
Network	IP header with source and destination IP, TTL, protocol
Data link	Frame header with source and destination MAC, plus a trailer with the FCS
Physical	Converts the frame into signals (bits)

Decapsulation is the reverse: each layer strips and inspects its own header and passes the payload up.

PDU (protocol data unit) is the name of the data unit at each layer: **bit** at physical, **frame** at data link, **packet** at network, **segment** at transport (datagram for UDP), and **message** or **data** above that.

Hierarchical layering is the arrangement itself: a stack of layers in which each layer offers services to the one above and consumes services from the one below, communicating logically with its peer layer in the other host through the headers it adds.

Physical layer: transmission media

Past-paper question

Sept 2026 CAE-1 Q2B. Describe the different transmission media used in computer networks. Explain guided and unguided media with their advantages, disadvantages, and applications.

Solution

Transmission media divide into **guided** (the signal is confined to a physical path) and **unguided** (the signal radiates through free space).

Guided media:

Medium	Characteristics	Use
Twisted pair (UTP/STP)	Two insulated copper wires twisted to cancel interference. UTP categories 5e, 6, 6a, 7. Cheap and easy to install; limited distance (≈100 m) and bandwidth.	LAN cabling, telephone lines.
Coaxial cable	A central conductor surrounded by a shield. Higher bandwidth and better shielding than twisted pair. Bulky and expensive.	Cable TV, older Ethernet (10Base2), broadband.
Optical fibre	Glass or plastic core carrying light. Single-mode for long haul, multimode for short. Enormous bandwidth, immune to electromagnetic interference, very secure, low attenuation; expensive and fragile.	Backbones, undersea cables, FTTH, data centres.

Unguided media:

Medium	Characteristics	Use
Radio waves	Omni-directional, penetrates walls, 3 kHz to 1 GHz.	AM/FM radio, Wi-Fi, mobile.
Microwaves	Line of sight, high frequency, needs dish alignment and towers; affected by rain.	Point-to-point links, satellite.
Infrared	Very short range, cannot pass through walls, needs line of sight.	TV remotes, short-range device links.

Choosing a medium comes down to four trade-offs: bandwidth needed, distance, immunity to interference, and cost. Fibre wins on all but cost and fragility, which is why it is used for the backbone and copper for the last hundred metres.

Switching

Past-paper question

Sept 2026 CAE-1 Q3A. Explain different switching techniques used in computer networks. Compare circuit switching and packet switching.

Solution

Switching is the technique by which a switched network moves data from source to destination through intermediate nodes.

Technique	How it works
Circuit switching	A dedicated physical path is established end to end before any data flows, and it is held for the whole session. Resources are reserved.
Message switching	The whole message is stored at each intermediate node, then forwarded. No dedicated path; store-and-forward. Obsolete.
Packet switching	The message is broken into packets that are forwarded independently, each carrying the destination address, sharing the links statistically.

Basis	Circuit switching	Packet switching
Path	Dedicated, established first	No dedicated path; each packet routed independently
Resource use	Reserved even when idle (wasteful)	Shared on demand (efficient)
Set-up delay	Yes, before data	None
Delay	Constant after set-up	Variable, due to queueing
Congestion	Refused at set-up if no capacity	Packets queue and may be dropped
Overhead	Minimal	Per-packet header
Best for	Voice, real-time streams	Burst data: web, email, file transfer
Example	PSTN	The Internet

Why the Internet uses packet switching: data traffic is bursty. Reserving a circuit for a user who is reading a web page wastes capacity that another user could use. Statistical multiplexing gives far higher utilisation at the cost of variable delay.

Multiplexing

Past-paper question

Sept 2026 CAE-1 Q4A and Oct 2025 Q1(a). What is multiplexing? Explain the need for multiplexing. Discuss FDM, TDM and WDM with suitable examples.

Solution

Multiplexing is the technique of combining several signals onto one high-capacity link so that it can be shared, and separating them again at the receiver (**demultiplexing**). The device is a **MUX**; the reverse is a **DEMUX**.

The need: a single link almost always has far more capacity than one user's traffic, and laying a separate cable per user is prohibitively expensive. Multiplexing converts one high-bandwidth link into many logical channels, so cost per user falls and utilisation rises.

Technique	How it separates users	Notes
FDM	Different **frequency** bands	Analogue; each user gets a slice of the spectrum continuously. Used in radio and TV broadcast, and ADSL.
TDM	Different **time slots**	Digital; users take turns on the whole link. Synchronous TDM assigns fixed slots (wasteful when a user is idle); statistical TDM assigns slots on demand.
WDM	Different **wavelengths** (colours) of light	The optical version of FDM, used in fibre backbones. DWDM packs dozens of wavelengths into one fibre.
CDM/CDMA	Different **codes**	All users transmit at once on the whole band; each uses a unique spreading code. Used in 3G mobile.

Remember

FDM vs TDM in one line each. FDM: every user transmits all the time, on a narrow band. TDM: every user uses the whole band, for a short slice of time. If the question asks for a difference, this sentence plus a diagram is a full answer.

The hierarchy: TDM carriers are built in steps. In the North American hierarchy, 24 DS-0 channels (64 kbps each) form a DS-1/T-1 at 1.544 Mbps; in the European hierarchy, 30 channels plus 2 for signalling form an E-1 at 2.048 Mbps.

Digital transmission: line coding and PCM

Past-paper question

Sept 2024 CAE-1 Q1. Explain the polar coding schemes of digital signals.

Solution

Line coding converts a sequence of bits into a digital signal. In **polar** schemes, two voltage levels are used, one positive and one negative, so the average DC level is zero.

Scheme	Rule	Property
NRZ-L	Level depends on the bit: one level for 0, the other for 1	Simple; long runs of the same bit cause loss of synchronisation
NRZ-I	Invert the level on a 1, hold it on a 0	Solves the long-run problem for 1s, not for 0s
Manchester	Transition in the middle of every bit; low-to-high for 0, high-to-low for 1	Self-clocking: the clock is recovered from the signal. Used in classic Ethernet
Differential Manchester	Always a transition in the middle; a transition at the start of the bit means 0, none means 1	Self-clocking, and polarity-independent
AMI	0 is zero volts; 1s alternate between + and −	Good DC balance; used in T1 lines

The cost of Manchester coding is that it needs twice the bandwidth of NRZ for the same bit rate, because every bit carries a transition. This is the standard trade-off: self-clocking costs bandwidth.

Past-paper question

Sept 2024 CAE-1 Q3. What is Pulse Code Modulation (PCM), and how does it convert analog signals into digital form for transmission?

Solution

PCM is the standard method of digitising an analogue signal. Three steps:

1. **Sampling.** Measure the analogue signal at regular intervals. By the **Nyquist theorem**, the sampling rate must be at least twice the highest frequency present: $f_s \geq 2f_{max}$. For voice band-limited to 4 kHz, $f_s = 8000$ samples per second.
2. **Quantisation.** Round each sample to the nearest of $L = 2^n$ discrete levels. The rounding error is **quantisation noise**; using more levels reduces it. Non-uniform quantisation (companding, as in μ-law) gives small signals finer resolution.
3. **Encoding.** Assign a binary code of n bits to each level.

Bit rate: $R = f_s \times n$ bits per second. For digitised voice, $f_s = 8000$ and $n = 8$, so $R = 64$ kbps. That 64

kbps is the DS-0 channel on which all TDM hierarchies are built.

Quantisation error is at most half a step: $\text{SNR}_{dB} \approx 6.02n + 1.76$ for an n-bit quantiser. Each extra bit buys about 6 dB.

Remember

Two capacity formulas. Nyquist for a noiseless channel: $C = 2B \log_2 L$ bps, where B is bandwidth and L the number of signal levels. Shannon for a noisy channel: $C = B \log_2(1 + S/N)$. Shannon is the hard ceiling; Nyquist is the ceiling when there is no noise.

Data link layer

Functions

Past-paper question

Oct 2025 Q1(a)(iii). Describe the concept of framing in data link control.

Solution

The data link layer turns the physical layer's raw bit stream into manageable units and manages the link. Its functions are:

- **Framing**: dividing the bit stream into frames, each with a header, payload and trailer.
- **Physical addressing**: source and destination MAC addresses.
- **Error control**: detect (and sometimes correct) errors with a trailer checksum.
- **Flow control**: prevent a fast sender from overwhelming a slow receiver.
- **Media access control**: decide who may transmit on a shared medium.

Framing is necessary because the physical layer delivers an undifferentiated stream of bits. The receiver must know where a frame begins and ends. Three techniques:

Technique	How
Character count	A field in the header states the frame length. Fragile: one corrupted count desynchronises everything.
Byte stuffing	A special flag byte marks the start and end; any flag byte occurring in the data is escaped with an extra byte.
Bit stuffing	A flag pattern (01111110) delimits frames; in the data, a 0 is inserted after any five consecutive 1s. HDLC uses this.

Error detection: parity, checksum and CRC

Simple and two-dimensional parity

A **single parity bit** makes the number of 1s even (even parity) or odd. It detects any *odd* number of bit errors, so it misses all two-bit errors.

Two-dimensional parity arranges data as a block and computes parity for each row *and* each column. It detects all single, double and triple errors, and can *correct* a single error by locating the failing row and column intersection.

Internet checksum

Send the 16-bit one's-complement sum of the data words as a checksum; the receiver adds all words including the checksum and expects all ones.

Exam Focus

Worked. Words 1001 1001 1001 1001 and 0101 0101 0101 0101.

Sum = 1110 1110 1110 1110; checksum = one's complement = 0001 0001 0001 0001.

Receiver adds all three: 1111 1111 1111 1111 — all ones, so no error is detected. A checksum is weaker than CRC; it is used in IP and UDP because it is cheap in software.

CRC — the calculation you will be asked to do

Past-paper question

Sept 2026 CAE-1 Q4B. Given the data bits 1101011011 and the generator bits 10011: (a) append the required number of zeros; (b) perform the modulo-2 binary division, showing all steps; (c) determine the CRC remainder and construct the final transmitted codeword.

Solution

The rule: if the generator has $n+1$ bits, append n zeros, divide modulo 2, and replace the zeros with the remainder.

Generator 10011 has 5 bits, so $n = 4$ and we append **4 zeros**.

Step 1 — append:

$$1101011011 + 0000 = 11010110110000$$

Step 2 — divide modulo 2 (XOR, no carries). Each time the leading bit is 1, XOR the generator into the window and drop the leading 0; when it is 0, shift one place.

Step	Current window
start	11010110110000
XOR at position 0	10011 ⇒ 01001110110000
shift to position 1	1001110110000
XOR at position 1	10011 ⇒ 0000010110000
shift to position 6	10110000
XOR at position 6	10011 ⇒ 0010100
shift to position 8	10100
XOR at position 8	10011 ⇒ 00111
remainder	1110

Let us verify the arithmetic rather than trust it. Position 0: 11010 ⊕ 10011 = 01001. Position 1: window 10011 ⊕ 10011 = 00000. What remains after the two leading XORs is the low bits 10110000. Position 6: 10110 ⊕ 10011 = 00101, leaving 0100. There is no 1 left at a position that can start a division except the last window 10100 at position 8: 10100 ⊕ 10011 = 00111. The remainder is 1110.

Step 3 — the codeword: replace the four appended zeros with the remainder 1110:

11010110111110

Step 4 — why it works. The receiver divides the received 14 bits by the same generator. If the remainder is 0000 the frame is accepted; anything else is discarded. This works because the transmitted polynomial is exactly divisible by $G(x)$.

Past-paper question

Oct 2025 CAE-II Q1A and Oct 2024 CAE-II Q1A (the same question in both papers). A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$. (a) What is the actual bit string transmitted? (b) Suppose the third bit from the left is inverted during transmission. How will the receiver detect this error?

Solution

Step 1 — the generator. $G(x) = x^3 + 1$ means coefficients at x^3 and x^0, so the bit pattern is 1001 (4 bits, degree 3). Append **3** zeros.

$$10011101 + 000 = 10011101000$$

Step 2 — divide:

Step	Window
start	10011101000
XOR at position 0	$1001 \Rightarrow 00001101000$
shift to position 4	1101000
XOR at position 4	$1001 \Rightarrow 0100000$
shift to position 5	100000
XOR at position 5	$1001 \Rightarrow 000100$
remainder	100

The remainder is 100, so the transmitted string is

10011101100

Step 3 — the inverted bit. The third bit from the left of 10011101100 is the 0 in position 3 (positions 1, 2, 3 are 1, 0, 0), so it becomes 1:

$$\text{received} = 10111101100$$

Divide the received bits by 1001. The remainder is 100, which is **not zero**, so the receiver discards the frame and requests a retransmission.

The point to write down: the receiver does not need to know *where* the error is or *what* it was. A non-zero remainder is enough. That is why CRC is the standard link-layer detector.

Do Not Confuse

Modulo-2 arithmetic means XOR, not subtraction. There are no borrows and no carries. If you find yourself writing a 2 or a −1 in the division, stop: you are doing ordinary binary subtraction.

Error correction: Hamming code

To correct a single error you need enough redundancy to identify *which* bit. With m data bits and r check bits, the condition is

$$2^r \geq m + r + 1.$$

For $m = 4$, $r = 3$ satisfies $8 \geq 8$, giving the classic (7,4) Hamming code. Check bits sit at positions 1, 2, 4, 8, …; each check bit covers the positions whose binary index contains that bit.

Remember

Detection versus correction. Detection only needs to answer "did something go wrong". Correction must answer "what exactly went wrong". Correction therefore needs far more redundancy, which is why links normally detect and retransmit (ARQ) rather than correct, and only real-time links (satellite, deep space) pay for forward error correction.

Data link control: flow control and ARQ

Past-paper question

Oct 2025 Q1(b)(i). Explain the working principle of the Go-Back-N ARQ protocol. Compare it with the Selective Repeat ARQ protocol in terms of efficiency.

Solution

Go-Back-N. The sender may have up to N unacknowledged frames outstanding. The receiver accepts only in-order frames and sends a cumulative ACK. If a frame is lost, the receiver discards everything after it, and the sender, on timeout, **retransmits that frame and all subsequent ones**.

Window size: $N = 2^m - 1$ for an m-bit sequence number.

Selective Repeat. Same windowing, but the receiver buffers out-of-order frames and ACKs each one individually; the sender retransmits **only** the missing frame.

Window size: $N = 2^{m-1}$.

Basis	Go-Back-N	Selective Repeat
Receiver buffer	None; out-of-order frames discarded	Buffers out-of-order frames
Retransmission	The lost frame and every frame after it	Only the lost frame
Window size	$2^m - 1$	2^{m-1}
Efficiency on a lossy link	Falls sharply as the error rate rises	Stays high
Complexity	Simple sender and receiver	More complex receiver and buffering

Efficiency. For a link with transmission time T_t and one-way propagation T_p, let $a = T_p/T_t$. The sender can keep the link busy only if

$$N \geq 1 + 2a.$$

The utilisation of Go-Back-N is $U = \dfrac{N}{1+2a}$ (capped at 1). For example with $a = 2$, a window of 5 gives $U = 5/5 = 1$, i.e. 100%; a window of 3 gives $U = 3/5 = 0.6$, i.e. 60%.

Selective Repeat reaches the same utilisation but wastes no capacity on retransmitting frames that arrived correctly. Go-Back-N is cheaper to implement; Selective Repeat is more efficient. That is the trade-off the question wants.

Past-paper question

Oct 2024 CAE-II Q1B. Explain the sliding window protocols in detail.

Solution

A **sliding window** bounds how many frames may be in flight. The sender keeps a window of at most N unacknowledged frames; each ACK slides the window forward. This gives flow control (the window never exceeds what the receiver can buffer) and efficiency (the sender does not stop and wait for each frame).

- Window size 1 with stop-and-wait: one frame at a time. Utilisation $= \dfrac{1}{1+2a}$, which collapses on links with a large propagation delay, such as satellite.
- Window size N: pipelining. Utilisation $= \dfrac{N}{1+2a}$, capped at 1.

The two ARQ implementations of the sliding window are Go-Back-N and Selective Repeat, compared above. **Piggybacking** is the optimisation in which a data frame travelling in the reverse direction carries the ACK for the forward direction, saving a separate ACK frame.

The MAC sublayer

Past-paper question

Sept 2026 CAE-1 Q2A. Explain the concept of Random Access Protocols used in the MAC sublayer. Discuss the working principles of ALOHA, Slotted ALOHA, CSMA, and CSMA/CD, highlighting their differences and limitations.

Solution

Random access means no station is assigned a turn in advance; any station may transmit when it has data, and a **contention resolution** procedure handles the resulting collisions. This is efficient for bursty traffic, which is why Ethernet uses it.

Protocol	Principle	Limitation
ALOHA	Transmit whenever you have data. If no ACK, wait a random time and retransmit.	Maximum throughput 18.4% of the channel; collisions are frequent
Slotted ALOHA	Time is divided into slots; transmit only at a slot boundary	Doubles efficiency to 36.8%; still collision-prone
CSMA	*Carrier sense*: listen before transmitting. Transmit only if the medium is idle	Collisions still possible because of propagation delay
CSMA/CD	CSMA plus *collision detection*: keep listening while transmitting, abort on collision, send a jam signal, back off exponentially	Only works on wired media where a station can hear while it talks
CSMA/CA	Collision *avoidance*: wait a random backoff before transmitting, and optionally use RTS/CTS to reserve the medium	Needed on wireless, where a station cannot detect collisions

Why throughput is 18.4% for ALOHA: in pure ALOHA a frame is vulnerable for two frame times, so the probability of no collision is e^{-2G} and the throughput $S = Ge^{-2G}$ peaks at $G = 0.5$, giving $S = 0.184$. For slotted ALOHA the vulnerable period halves, so $S = Ge^{-G}$ peaks at $G = 1$ with $S = 0.368$.

Past-paper question

Oct 2025 CAE-II Q1B. A LAN has 5 computers (A, B, C, D, E) connected using CSMA/CD. The time it takes a signal to travel between any two computers is 25 μs. Each frame takes 100 μs to send. Computer A starts sending a frame at the same time as computer C. (a) What happens when A and C send at the same time? (b) How do the computers know that a collision has happened? (c) What do the computers do after detecting the collision?

Solution

(a) What happens. Both stations sense the medium idle and begin transmitting simultaneously. Their signals meet at the midpoint of the path between them, about 12.5 μs after both started, and the overlapping signals corrupt each other. Neither frame is delivered.

(b) How they know. A station transmitting with CSMA/CD continues to monitor the medium. When it senses a signal level or waveform different from its own, it concludes a collision has occurred. The time to notice is at most one round trip, $2T_p = 2 \times 25 = 50$ μs.

The key condition. CSMA/CD only works if the station is *still transmitting* when it learns of the collision, that is

$$T_t \geq 2T_p.$$

Here $T_t = 100$ μs and $2T_p = 50$ μs, so the condition holds comfortably: the station is only halfway through its frame when it detects the collision. If the frame were shorter than 50 μs, the station would finish and believe it had succeeded.

(c) What they do next. Each station aborts its transmission immediately and sends a short **jam signal** (32 to 48 bits) so that every other station also notices the collision. Then both enter **binary exponential backoff**: after the n-th collision a station waits a random number of *slot times* ($2T_p = 50\ \mu$s) chosen from $\{0, 1, \ldots, 2^n - 1\}$, and tries again. After 16 failed attempts the frame is abandoned and reported as an error.

Efficiency of this LAN: $U = \dfrac{T_t}{T_t + 2T_p} = \dfrac{100}{100 + 50} = 0.667$, that is 66.7%.

Remember

CSMA/CD versus CSMA/CA. Wired Ethernet can detect a collision because the transmitter can listen to the same cable it is transmitting on. A wireless station cannot: its own transmission drowns out everything else (the "hidden node" problem), so it cannot detect a collision and must *avoid* one instead, using random backoff and optional RTS/CTS. That single physical fact is the whole reason 802.3 uses CD and 802.11 uses CA.

Answer skeletons

Answer Skeleton

Define-and-list (3 to 10 marks). Give a one-sentence definition, then a labelled list. For the five components of data communication, name all five in order and add a clause each. Never leave a list unlabelled.

Answer Skeleton

Compare-two-things (10 marks). A table, at least five rows, each row a single basis of comparison. OSI versus TCP/IP; circuit versus packet; FDM versus TDM; GBN versus SR; CSMA/CD versus CSMA/CA; guided versus unguided media. Write the table first, then one sentence of conclusion.

Answer Skeleton

CRC calculation (10 marks). State the generator in bits, state how many zeros to append and why, show the division step by step, box the remainder, then write the codeword. The steps carry the marks, not the answer.

Answer Skeleton

Explain-a-mechanism (10 marks). Definition, then the mechanism in numbered steps, then a diagram, then one advantage and one limitation. For a protocol this is: what it does, how it does it, when it fails.

Self-test

1. Define data communication and list its five components.
2. Differentiate between bandwidth, data rate and throughput.
3. What is jitter and why does it matter for voice?
4. State the seven OSI layers with their PDUs.
5. Compare OSI and TCP/IP in a table.
6. What is encapsulation? Name the PDU at each layer.
7. Distinguish guided from unguided media with two examples each.
8. Compare circuit switching and packet switching.
9. What is the need for multiplexing? Differentiate FDM and TDM.
10. What is PCM? State the three steps and the bit rate for digitised voice.
11. State the Nyquist and Shannon capacity formulas.
12. What is framing? Name three framing techniques.
13. Perform the CRC for data 10011101 with generator $x^3 + 1$.
14. How many check bits are needed to correct a single error in 4 data bits?
15. State the Go-Back-N and Selective Repeat window sizes for a 3-bit sequence number.

16. Derive the condition for CSMA/CD to work.
17. Why is the maximum throughput of ALOHA 18.4%?
18. Why does wireless use CSMA/CA instead of CSMA/CD?

Answers

1. The exchange of data between devices over a medium. Components: message, sender, receiver, transmission medium, protocol.
2. Bandwidth is the capacity of the medium (Hz or max bit rate); data rate is the actual bit rate of a device; throughput is the successful delivery rate after overhead, always $\leq$ bandwidth.
3. Jitter is the variation in latency between successive packets. Real-time audio and video need steady delay, so jitter causes glitches even when average latency is acceptable.
4. Physical (bit), data link (frame), network (packet), transport (segment), session, presentation, application (data).
5. See the comparison table: 7 layers versus 4; theoretical versus practical; session and presentation folded in; TCP/IP is what is deployed.
6. Adding a header at each layer on the way down. PDUs: bit, frame, packet, segment, data.
7. Guided confines the signal: twisted pair, coaxial, fibre. Unguided radiates: radio, microwave, infrared.
8. Circuit switching reserves a dedicated path and is good for voice; packet switching forwards independent packets and is good for bursty data.
9. Multiplexing shares one link among many users to cut cost and raise utilisation. FDM separates by frequency band; TDM separates by time slot.
10. PCM digitises analogue signals by sampling, quantising and encoding. For voice, $f_s = 8000$ and $n = 8$, giving 64 kbps.
11. Nyquist $C = 2B \log_2 L$; Shannon $C = B \log_2(1 + S/N)$.
12. Framing divides the bit stream into frames. Techniques: character count, byte stuffing, bit stuffing.
13. Generator `1001`, append 3 zeros, remainder `100`, codeword `10011101100`.
14. $r = 3$, since $2^3 = 8 \geq 4 + 3 + 1 = 8$.
15. GBN: $2^3 - 1 = 7$. Selective Repeat: $2^{3-1} = 4$.
16. $T_t \geq 2T_p$: the frame must take at least as long as the round trip, otherwise the sender finishes before learning of the collision.
17. A frame is vulnerable for two frame times, so $S = Ge^{-2G}$, which peaks at $G = 0.5$ with $S = 0.184$.
18. A wireless station cannot hear collisions while transmitting because its own signal dominates, so it must avoid them by backoff and RTS/CTS.

Last-minute page

DATA COMM
5 components: message, sender, receiver, medium, protocol

Bandwidth $>$ throughput always
Latency vs jitter

OSI 7
Physical, Data link, Network, Transport, Session, Presentation, Application
PDU: bit, frame, packet, segment

TCP/IP 4
Application, Transport, Internet, Network access

MEDIA
Guided: twisted pair, coax, fibre
Unguided: radio, microwave, IR

SWITCHING
Circuit: reserved path, voice
Packet: independent, data
Message: store and forward

MULTIPLEXING
FDM: frequency, all the time
TDM: time slot, whole band
WDM: wavelengths, fibre

PCM
Sample $\rightarrow$ quantise $\rightarrow$ encode
$f_s \geq 2f_{max}$
8000 $\times$ 8 = 64 kbps

CAPACITY
Nyquist $2B \log_2 L$

Shannon $B \log_2(1 + S/N)$

ERROR CONTROL
Parity: odd/even, misses 2-bit
Checksum: 1s complement sum
CRC: XOR division, remainder

ARQ
GBN window $2^m - 1$
SR window 2^{m-1}
$N \geq 1 + 2a$ for full use

MAC
ALOHA 18.4%, slotted 36.8%
CSMA/CD: wired, $T_t \geq 2T_p$
CSMA/CA: wireless, backoff

DO NOT CONFUSE
Bandwidth vs throughput
Latency vs jitter
FDM (frequency) vs TDM (time)
CSMA/CD (detect) vs CSMA/CA (avoid)

Module 1 is definition-and-table territory. Learn the five components, the seven layers with PDUs, and the three comparisons: OSI/TCP-IP, circuit/packet, FDM/TDM.